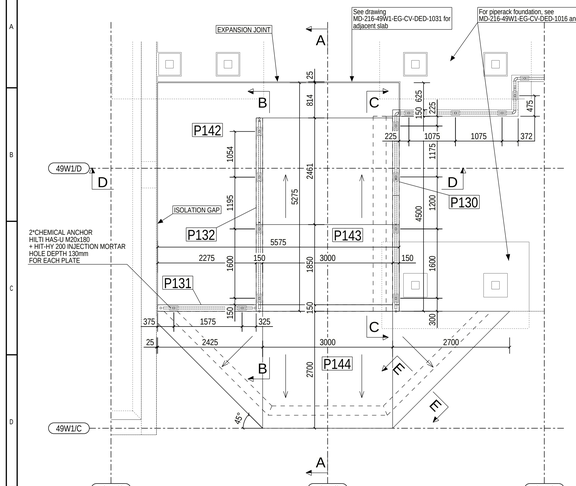
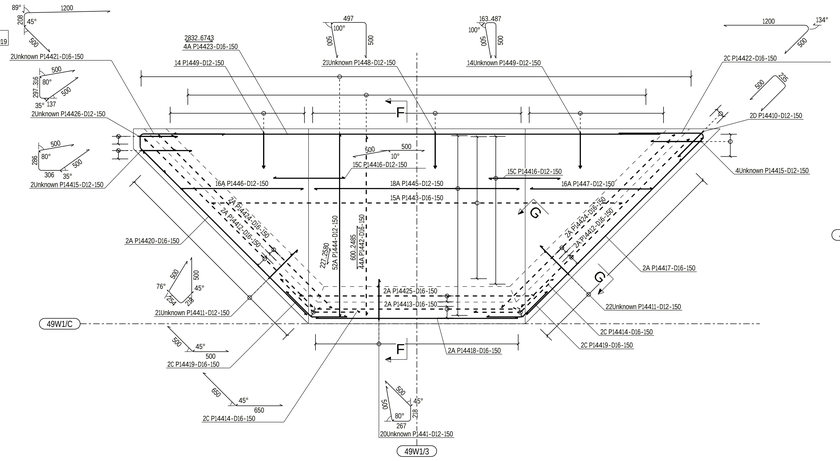
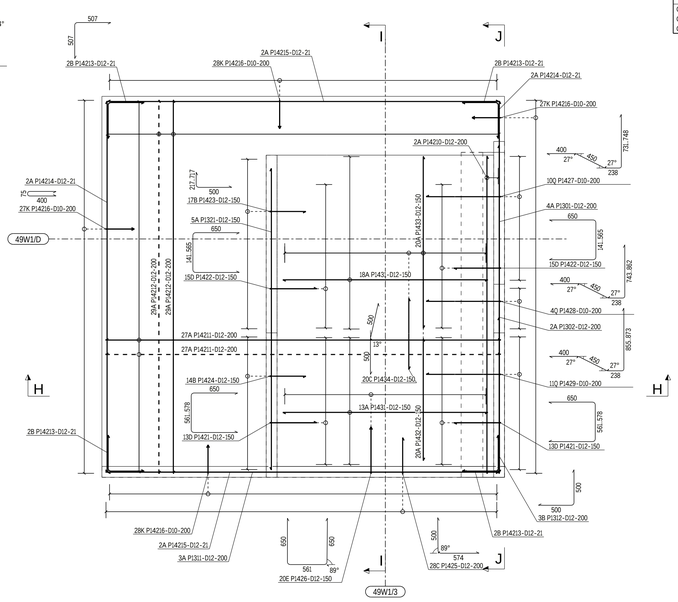
MD-216-49WL-EG-CV-DED-1032 | REV D01 | DRAWING NUMBER

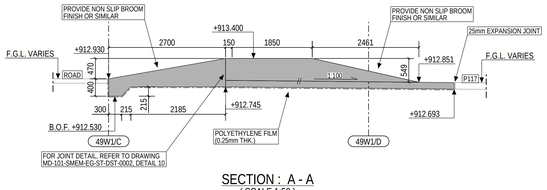


R.C.C RAMP AND GROUND SLAB DIMENSIONS
(SCALE 1:50)

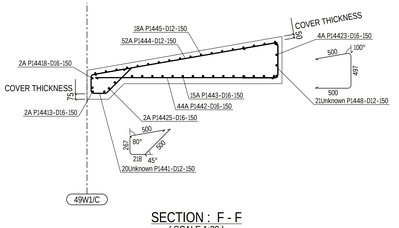


R.C.C RAMP REINFORCEMENT
(SCALE 1:30)

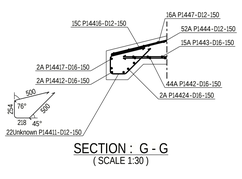




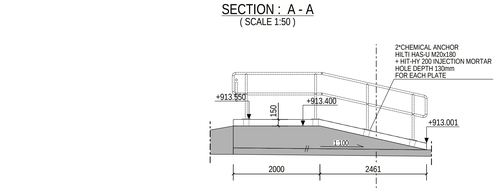
SECTION : A - A
(SCALE 1:50)



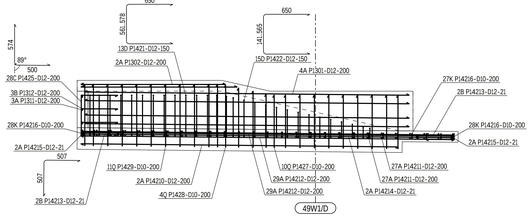
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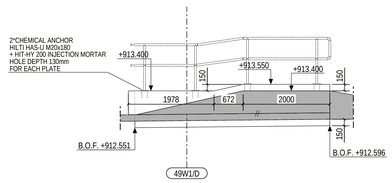
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(SCALE 1:30)



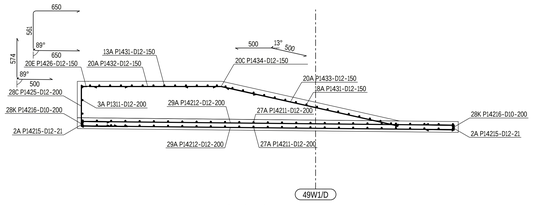
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(SCALE 1:50)



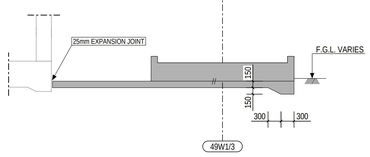
SECTION : J - J
(SCALE 1:30)



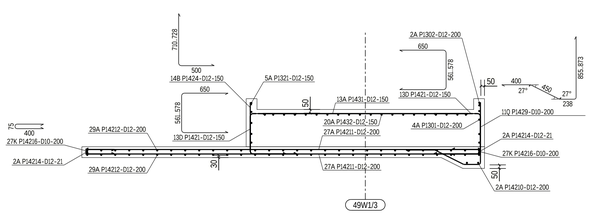
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(SCALE 1:50)



SECTION : I - I
(SCALE 1:30)



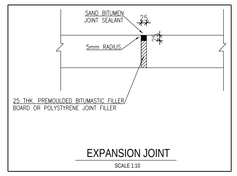
SECTION : D - D
(SCALE 1:50)



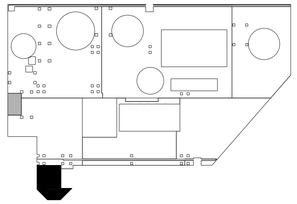
SECTION : H - H
(SCALE 1:30)

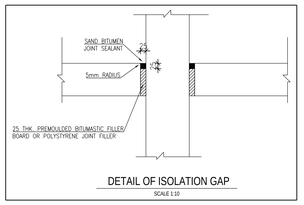
SECTION : E - E
(SCALE 1:50)

R.C.C RAMP AND GROUND SLAB REINFORCEMENT
(SCALE 1:30)



EXPANSION JOINT
SCALE 1:10



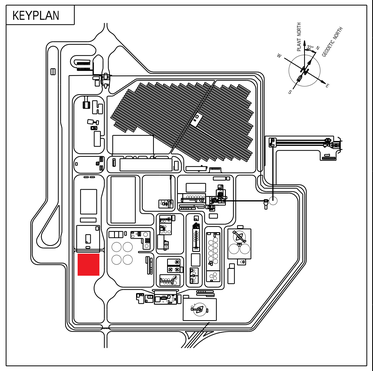


DETAIL OF ISOLATION GAP
SCALE 1:10

LEGEND:
R.L - REDUCED LEVEL
T.O.C - TOP OF CONCRETE
B.O.F. - BOTTOM OF FOUNDATION
F.G.L. - FINISHED GROUND LEVEL
P.C.C. - PLAIN CEMENT CONCRETE
TYP - TYPICAL
MKD - MARKED
CL - CENTER LINE
BOTTOM REINF. (B)
TOP REINF. (T)
SBC - SAFE BEARING CAPACITY
MIN. - MINIMUM
REF. - REFERED
GA - GENERAL ARRANGEMENT
DWG. - DRAWING
R.C.C. - REINFORCED CEMENT CONCRETE

NOTES:
1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS ARE IN METRES UNLESS OTHERWISE STATED
2. GRADE OF CONCRETE SHALL BE C35 (35MPa) UNLESS NOTED OTHERWISE & C15 (15MPa) FOR BUNDING CONCRETE.
3. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR ASTM A615-GRADE 60 (fy=420MPa)
4. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE AS FOLLOWS:
a) FOUNDATION: 75 mm (BOTTOM & SIDE)
b) FOUNDATION: 50 mm (TOP)
c) COLUMN & PEDESTAL: 50 mm
5. MINIMUM LAP LENGTHS OF REINFORCEMENT SHALL BE 40XBAR DIAMETER
6. FOR STANDARD REINFORCEMENT DETAILS & GENERAL NOTES FOR CIVIL WORKS REFER TO DRAWING NUMBER MD-216-1000-EG-CV-DED-1005 & 1001
7. ANCHOR BOLTS AND EMBEDDED STEELWORK SHALL BE HOT DIP GALVANIZED
8. T.O.C. & TOP OF SCREED TOLERANCE SHALL BE WITHIN +/- 10MM OF DESIGN LEVEL
9. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF FOUNDATION IS CONSIDERED AS 375 KN/SQ.M.
10. REBARS CAN BE MOVED LOCALLY BY +/- 25mm TO AVOID INTERFERENCE WITH EMBEDDED PARTS

KEYPLAN



CONCRETE INFORMATION

Concretes used	C35	
Concrete cover 1	75±10mm, 50±10mm	BOTTOM OF RAMP, SIDES AND TOP OF RAMP
Concrete cover 2	30±10mm	GROUND SLAB

REINFORCING BAR LIST — CAST UNIT CODE: P130 — PCS: 1

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	BENDING DIMENSIONS [mm] a	b	c	d	e	u	v	x	TD	COMMENT
A	P1301	4	B500B	12	4500		16.0	4500									
A	P1302	2	B500B	12	2160		3.7	2160									
3 AT GROUP		2	PCS	...	2000	160	...	2005									

REINFORCING BAR TOTAL WEIGHT [kg] 19.7

REINFORCING BAR LIST — CAST UNIT CODE: P131 — PCS: 1

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	BENDING DIMENSIONS [mm] a	b	c	d	e	u	v	x	TD	COMMENT
A	P1311	3	B500B	12	5420		14.5	5425									
B	P1312	3	B500B	12	970		2.6	500	500				90			60	

REINFORCING BAR TOTAL WEIGHT [kg] 17.0

REINFORCING BAR LIST — CAST UNIT CODE: P132 — PCS: 1

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	BENDING DIMENSIONS [mm] a	b	c	d	e	u	v	x	TD	COMMENT
A	P1321	5	B500B	12	4310		14.9	4311									
2 AT GROUP		5	PCS	...	2130	545	...	2132									

REINFORCING BAR TOTAL WEIGHT [kg] 14.9

REINFORCING BAR LIST — CAST UNIT CODE: P142 — PCS: 1

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	BENDING DIMENSIONS [mm] a	b	c	d	e	u	v	x	TD	COMMENT
D	P1421	26	B500B	12	1810		41.8	650	578	650						60	
2 AT GROUP		13	PCS	...	1800	1	...	650	561	650							
D	P1422	30	B500B	12	1800		42.5	650	565	650						60	
2 AT GROUP		15	PCS	...	1380	30	...	650	141	650							
B	P1423	17	B500B	12	1180		14.1	717	500				90			60	
1 AT GROUP		17	PCS	...	680	31	...	217	500				90				
B	P1424	14	B500B	12	1190		14.8	728	500				90			60	
1 AT GROUP		14	PCS	...	1180	1	...	710	500				90				
C	P1425	28	B500B	12	1040		26.0	574	500				89			60	
E	P1426	20	B500B	12	1800		32.0	650	561	650			89			60	
Q	P1427	10	B500B	10	1800		11.1	716	198	434	392		27	212	40		
5 AT GROUP		10	PCS	...	1790	1	...	699	198	434	392		27	212			
Q	P1428	4	B500B	10	1920		4.6	830	198	434	392		27	212	40		
1 AT GROUP		4	PCS	...	1800	40	...	711	198	434	392		27	212			
Q	P1429	11	B500B	10	1930		13.1	841	198	434	392		27	212	40		
1 AT GROUP		11	PCS	...	1910	2	...	823	198	434	392		27	212			
A	P14210	2	B500B	12	4400		7.8	4400									
A	P14211	54	B500B	12	5470		262.5	5475									
A	P14212	58	B500B	12	5170		266.6	5176									
B	P14213	8	B500B	12	980		7.0	507	507				90			60	
A	P14214	4	B500B	12	5140		18.3	5146									
A	P14215	4	B500B	12	5410		19.2	5417									
K	P14216	110	B500B	10	820		56.2	400	400							40	

REINFORCING BAR TOTAL WEIGHT [kg] 837.7

REINFORCING BAR LIST — CAST UNIT CODE: P143 — PCS: 1

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	BENDING DIMENSIONS [mm] a	b	c	d	e	u	v	x	TD	COMMENT
A	P1431	31	B500B	12	2840		78.2	2840									
A	P1432	20	B500B	12	1690		30.0	1690									
A	P1433	20	B500B	12	2440		43.4	2441									
C	P1434	20	B500B	12	990		17.7	500	500				13			60	

REINFORCING BAR TOTAL WEIGHT [kg] 169.3

REINFORCING BAR LIST — CAST UNIT CODE: P144 — PCS: 1

TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	BENDING DIMENSIONS [mm] a	b	c	d	e	u	v	x	TD	COMMENT
**	P1441	20	B500B	12	1420		25.3										
A	P1442	44	B500B	16	2480		131.7	2485									
B AT GROUP		44	PCS	...	600	44	...	600									
A	P1443	15	B500B	16	7520		131.7	7524									
1 AT GROUP		15	PCS	...	3590	281	...	3592									
A	P1444	52	B500B	12	2590		84.9	2590									
1 AT GROUP		52	PCS	...	220	46	...	227									
A	P1445	18	B500B	12	2840		45.4	2840									
A	P1446	16	B500B	12	2300		19.2	2302									
2 AT GROUP		16	PCS	...	240	137	...	247									
A	P1447	16	B500B	12	2320		19.2	2326									
1 AT GROUP		16	PCS	...	240	139	...	247									
**	P1448	21	B500B	12	1440		26.9										
**	P1449	28	B500B	12	1430		31.6										
3 AT GROUP		14	PCS	...	1110	25	...										
D	P14410	2	B500B	12	1210		2.1	500	270	500						60	
1 AT GROUP		2	PCS	...	1170	40	...	500	235	500							
**	P14411	43	B500B	12	1410		54.1										
A	P14412	4	B500B	16	3290		20.8	3294									
A	P14413	2	B500B	16	2750		8.7	2751									
C	P14414	4	B500B	16	1290		8.2	650	650				45			80	
**	P14415	6	B500B	12	1530		8.2										
C	P14416	30	B500B	12	990		26.6	500	500				10			60	
A	P14417	2	B500B	16	3460		11.0	3468									
A	P14418	2	B500B	16	2800		8.8	2800									
C	P14419	4	B500B	16	990		6.3	500	500				45			80	
A	P14420	2	B500B	16	3210		10.2	3214									
**	P14421	2	B500B	16	1850		5.9										
C	P14422	2	B500B	16	1690		5.4	500	1200				134			80	
A	P14423	4	B500B	16	6740		34.1	6743									
26 AT GROUP		4	PCS	...	2830	***	...	2832									
A	P14424	4	B500B	16	3390		21.5	3394									
A	P14425	2	B500B	16	3250		10.3	3251									
**	P14426	2	B500B	12	1480		2.6										
4 AT GROUP		2	PCS	...	1390	90	...										

REINFORCING BAR TOTAL WEIGHT [kg] 760.5

REINFORCEMENT BENDING SHAPES:



DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-101-SMEM-EG-ST-DST-0001	CONCRETE STUCTURAL STANDARDS
MD-216-49WL-EG-CV-GAD-1003	1-1D/A-E GENERAL ARRANGEMENT DRAWING
MD-216-49WL-EG-CV-GAD-1001	1-B/E-J GENERAL ARRANGEMENT DRAWING

DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-1000-EG-ST-CRT-1001	DESIGN CRITERIA
MD-216-49WL-EG-CV-GAD-1002	7-12/D-J GENERAL ARRANGEMENT DRAWING

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
D01	29.01.2024	AS-BUILT	JonRen	MauNot	NamWu	LuiPz	
C01	28.02.2022	ISSUED FOR CONSTRUCTION	JonRen	MauNot	NamWu	LuiPz	
A01	08.02.2022	ISSUED FOR APPROVAL	JonRen	MauNot	NamWu	LuiPz	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO DRAWING NUMBER: XXXXX — REVISION: 2

OUTOTEC DRAWING NUMBER: 900707_GAU01_AG11_CTB01_49032 — DOCUMENT ID: 0U100745539 — REVISION: 2

LARSEN & TOUBRO

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DRAWN	JonRen	08.02.2022
CHECKED	MauNot	08.02.2022
ENGINEER	NamWu	08.02.2022
APPROVED	LuiPz	08.02.2022
EMPLOYER		
SCALE	A0	

MANSOURAH - MASSARAH GOLD PROJECT
49W1 RAW WATER TREATMENT PLANT
SOUTHERN RAMP
DETAIL- AND REINFORCEMENT DRAWING

DRAWING NUMBER: MD-216-49WL-EG-CV-DED-1032 — REV: D01